LLM-Primary Gold Annotation for Cyber Threat Intelligence Named Entity Recognition: Independent Evaluation of Token-Classification Architectures*

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Abstract

Cyber threat intelligence (CTI) reports contain entity types with no general-domain analogue: ATT&CK technique identifiers, file hashes, TLP markings, threat actor aliases. Named entity recognition (NER) models handle these poorly. Prior work in this space commonly evaluates against *silver* labels produced by the same class of dictionary/rule annotator being improved upon. However, our experiments showed that this evaluation is circular: a checkpoint that scores 0.74 F1 against dictionary-derived silver labels falls to 0.35 against independently produced gold, because the silver labels share the model's blind spots rather than measuring against them. In this work, we instead generate gold labels directly with a frontier LLM (Claude Sonnet 5), which reads full advisory text and proposes exhaustive entity annotations under per-label span-format validators; a human curates the evaluation split. On this gold we fine-tune three token-classification encoders: RoBERTa-base, RoBERTa-large [8], and SecureBERT [1], a RoBERTa-architecture model domain-pretrained on security text. We evaluate all three with an identical document-level harness, alongside GLiNER [11], a zero-shot span-matching architecture, fine-tuned on the identical gold. SecureBERT performs best (0.71 strict micro-F1), ahead of RoBERTa-base (0.66) and RoBERTa-large (0.65), indicating that domain-adaptive pretraining matters more than parameter count on this task. GLiNER, once retrained on the same independent gold, is the weakest of the four (0.56). All three token classifiers approach ceiling on pattern-structured types (attack technique IDs, CVE identifiers, TLP markings, email addresses) and remain substantially weaker on open-vocabulary semantic types (threat actor group, hacking tool). Extending the comparison to a true zero-shot frontier LLM (Gemini 3.5 Flash) on the same gold, the frontier model surpasses every fine-tuned local model on F1 (0.795 vs. SecureBERT's 0.712), at a cost of 94.5 s mean latency per document and metered per-token API pricing, against SecureBERT's 1.8 s, free, fully offline inference: a 51× speed difference for 8 points of F1. GLiNER, although local, does not share that speed advantage: at 59.7 s per document it is only 1.6× faster than the cloud API and is dominated by SecureBERT on both accuracy and latency. We report these trade-offs directly; the appropriate operating point depends on deployment context rather than on either a "larger models win" or a "local models win" generalization.

Keywords

named entity recognition, cyber threat intelligence, LLM-based annotation, evaluation methodology, domain-adaptive pretraining

1 Introduction

Cyber threat intelligence reports (government advisories, malware analyses, incident write-ups) are the primary vehicle by which defenders share knowledge, yet extracting structured information from them remains largely manual. Named entity recognition over CTI text must handle entity types with no general-domain analogue: ATT&CK technique identifiers, file hashes, TLP markings, YARA rule names, threat actor aliases. Human annotation at scale is expensive in this domain because it requires security expertise. A common response, and the baseline strategy this work replaces, is to avoid human annotation entirely: build a *silver*-labeled corpus from agreement between a dictionary/regex annotator (STIX indicators, MITRE ATT&CK lookups) and zero-shot GLiNER [11] predictions, and evaluate against held-out silver labels produced by the same dictionary annotator.

That evaluation is circular, and the gap it hides is large. A blind human audit of predicted spans (Section 4.1) found the dictionary annotator's own gold-recall to be materially incomplete (half of the spans the silver evaluation scored as model false positives were correct entities under direct human judgment), so a model that learns to reproduce the dictionary's blind spots is rewarded, not penalized, by silver-label scoring. We confirm the effect end to end: rescoring a dictionary-silver-trained checkpoint against independently produced gold drops its score from 0.74 to 0.35 F1 (Section 4). The drop reflects a change in the evaluation reference, not a change in the checkpoints themselves.

This finding motivates the pipeline described here. Rather than avoiding human involvement entirely, we move it to where it is cheapest to apply correctly: a frontier LLM (Claude Sonnet 5) reads each advisory's full text and proposes exhaustive gold-standard entity annotations, subject to per-label span-format validators at generation time; a human then curates the evaluation split (accepting, rejecting, or relabeling proposals), while the larger training split remains LLM-only. This approach is not free of human effort, but it replaces exhaustive first-pass annotation with targeted review; critically, it produces evaluation gold that is *independent* of any dictionary or rule system a model might learn to imitate.

We ask four research questions:

RQ1 How large is the gap between circular (dictionary-silver) evaluation and independent (LLM-primary, human-curated)

*Code, data, and fine-tuned models are publicly available: https://github.com/dandye/nlp_capstone. See Section 8.

evaluation for a model trained on the old weak-supervision pipeline?

RQ2 Once both training and evaluation labels come from the independent LLM-gold pipeline, does model scale (RoBERTa-base vs. RoBERTa-large) or domain-adaptive pretraining (SecureBERT) matter more for token-classification CTI NER?

RQ3 Which entity types remain hard even after retraining on independent gold, and does the pattern track training-example frequency, structured-vs-semantic label type, or both?

RQ4 How does this set of fine-tuned token classifiers compare to a zero-shot span-matching architecture (GLiNER) retrained on the identical independent gold, and to a frontier LLM run zero-shot?

All four research questions are answered with data trained and scored on the current LLM-gold pipeline: RQ4's architecture half with a GLiNER checkpoint retrained on the identical gold, and its frontier-LLM half with a zero-shot Gemini 3.5 Flash run on the same gold and harness (Section 4.4). The prior iteration's Gemini numbers, scored against dictionary-silver gold, are superseded and not reported.

2 Related Work

GLiNER [11] treats NER as matching natural-language entity-type descriptions against learned span embeddings, enabling zero-shot extraction of arbitrary types; its schema-driven successor GLiNER2 [10] unifies NER with classification and structured extraction in one model. Distantly supervised NER was formalized for open-domain text by BOND [7], which fine-tunes a BERT-based token classifier [4] on knowledge-base-derived labels and then denoises them through a teacher–student self-training loop. The failure modes BOND targets (incomplete and noisy weak labels) are the same failure modes that motivate this work's move away from dictionary supervision.

Cybersecurity-specific NER corpora suffer acutely from schema fragmentation: Schmidt et al. [9] show that naively unifying four CTI datasets under a coarse shared label set *hurts* performance relative to single-dataset training, while Ech-Chammakhy et al. [5] demonstrate that principled, STIX-anchored harmonization recovers a 30% relative F1 gain over the same naive union: evidence that *how* labels are unified matters as much as how much data is unified. Our LLM-primary annotation pipeline sidesteps unification entirely by defining the label schema once and generating gold directly for it, rather than merging pre-existing dictionary schemas. Deka et al. [3] extend the entity-type inventory to attribution-specific concepts (impact, course-of-action) with deliberately long, sentence-level spans, and find instruction-fine-tuned LLMs outperform encoder taggers on their benchmark, motivating our use of an LLM as the *annotator* rather than the deployed extractor, although Section 4.4 finds the same LLM-outperforms-encoder pattern holds even for a deployed zero-shot extractor on our corpus, so that choice is ultimately one of deployment cost rather than of where an LLM can add value. Alam et al. [2] benchmark LLMs directly on CTI tasks including entity extraction, providing the frontier-model baseline methodology that RQ4 adapts.

LLM-as-annotator is itself an active line of work distinct from LLM-as-extractor: rather than running a frontier model zero-shot at inference time, we use it once, offline, to produce training and evaluation labels for smaller, cheaper, deployable encoders: a distillation-flavored setup, except that the teacher's output is explicitly checked (span-format validators, human curation) rather than trusted directly. The failure mode being guarded against is exactly the one BOND and the dictionary-silver approach share: an annotator's systematic blind spots being learned uncritically by the student.

3 Data and Method

3.1 Pipeline

We scrape approximately 300 CISA cybersecurity advisories; each is converted from PDF to markdown via docling, then stripped of layout artifacts and normalized (page-furniture deduplication, pagination removal, split-hash rejoining) to a clean, model-facing text version. For 82 of these advisories we generate gold entity annotations across 16 CTI-specific labels[1] directly with a frontier LLM (Claude Sonnet 5): the model reads each advisory's full clean text and proposes exhaustive entity spans, which are verbatim-matched back to the source text and rejected if they do not match exactly, and further gated by per-label span-format validators (hash length and character set, CVE/CWE/ATT&CK technique-ID grammar, IP/email format including defanged forms, TLP enumeration) so that structurally malformed proposals never enter the gold set. This is an annotation-time generation gate, not a post-hoc filter.

The 66 training-split advisories receive LLM annotation only. The 16 validation/test advisories additionally pass through human curation via a purpose-built review tool: a human accepts, rejects, or relabels each LLM proposal and may add entities the LLM missed, including an `UNK` category (not used for training or evaluation directly) for real entities that fit no current label, which we use to identify candidate label types. Evaluation gold is therefore human-verified; training gold is not. We report this as a limitation (Section 5) but consider it acceptable because the LLM-primary approach's failure mode (proposing an entity a human would reject) is structurally different from the dictionary approach's failure mode of *never proposing* entire classes of entity for which it has no rule, and a human-verified evaluation set is positioned to catch either.

The training split contains 6,701 gold entities across 66 advisories; the evaluation split (8 validation, 8 test advisories) contains 1,995 gold spans in the test set alone once chunked for scoring (Section 3.4). This is roughly 3.5× the raw entity volume of the dictionary-silver baseline's test set on the same page count, because dictionary/regex annotation structurally omits several semantic label types (e.g., *affected software product*, *hacking tool*) that an LLM reader identifies from context.

3.2 Split

We split by advisory *document*, not by chunk, before chunking for training (66 advisories train / 8 validation / 8 test) so that overlapping text windows from the same document cannot appear on both sides of the split. This is a deliberate fix carried over from

[1] affected software product, attack technique id, attack technique name, cve identifier, cwe identifier, email address, file hash, file name, file path, hacking tool, ip address, malware family, tactic, threat actor group, tlp classification, yara rule name.

the baseline pipeline: chunk-level splitting leaks document context (shared boilerplate, repeated entity mentions) across the train/test boundary and inflates apparent performance.

3.3 Models

RoBERTa-base and RoBERTa-large [8]. Standard BIO token classifiers fine-tuned on the LLM-gold training split. Because gold spans are word-level and RoBERTa's tokenizer is subword-based, we tag the first wordpiece of each word and mask continuation wordpieces from the loss. Corpus chunks vary widely in length, and long or hash-heavy chunks fragment into more wordpieces than a single window admits under naive truncation; we instead tokenize with overlapping sliding windows (a 510-wordpiece budget per window, 50-word overlap between windows) so that every word is covered by at least one window, and take each word's predicted tag from the window in which it sits most centrally at inference time. RoBERTa-base trains for 8 epochs at learning rate 2e-5, effective batch size 16; RoBERTa-large uses the identical recipe with effective batch size 32 (gradient accumulation, to fit the larger model). Length-grouped batching reduces padding waste across the wide chunk-length range.

SecureBERT [1]. The same RoBERTa architecture and fine-tuning recipe as RoBERTa-base (8 epochs, learning rate 2e-5, effective batch size 16), but initialized from a checkpoint further pretrained on a security-text corpus rather than general-domain text. Because architecture and fine-tuning recipe are held identical to RoBERTa-base, any difference between the two isolates the effect of domain-adaptive pretraining from the effect of the fine-tuning data or procedure.

GLiNER [11]. `urchade/gliner_medium-v2.1`, zero-shot span-matching, fine-tuned on the identical LLM-gold training split: 5 epochs, learning rate 5e-5, effective batch size 32 (batch 8, gradient accumulation 4), best checkpoint selected by validation loss (GLiNER's trainer reports no span-level F1 during training; that is computed separately with the same harness used for the token classifiers). Unlike the token classifiers, GLiNER handles its own tokenization and length limits internally rather than using the sliding-window scheme of Section 3.4; several of the longest test-set chunks were truncated at its 384-token limit, which we record as a limitation (Section 5) because it plausibly disadvantages GLiNER specifically on labels that skew toward the end of long chunks.

Frontier LLM.. Gemini 3.5 Flash, run zero-shot with no fine-tuning, following Alam et al. [2]. Full documents are covered by overlapping 15,000-character windows (500-character overlap) so that no input is truncated, in contrast to GLiNER's fixed token limit above; per-window entity proposals are merged by exact text-and-label match before scoring, since a single entity mention can legitimately recur across overlap regions. Reasoning is disabled (`thinking_budget=0`): NER requires no reasoning tokens, and hidden reasoning output made early informal runs both slow and expensive.

3.4 Evaluation

All models are scored with the same document-level harness (`ner.evaluate`): predictions from overlapping chunks are merged back to document-global token indices before matching, so that entities in chunk-overlap regions are not double-counted. We report strict (exact span-and-label match) and relaxed (label match plus span overlap) micro/macro precision, recall, and F1 on the 8-document test split, using the identical scoring implementation across models so that the comparison in Section 4 is not an artifact of different matching rules.

4 Results

4.1 RQ1: Circular vs. Independent Evaluation

Table 1 rescores the baseline pipeline's checkpoints (trained on dictionary-silver labels, previously reported at 0.74 and 0.76 F1) against the current, independently produced LLM+human gold, on the same 8 test advisories (1,995 gold spans) used throughout this paper. Both token-classification checkpoints collapse: RoBERTa-base from 0.740 to 0.351, SecureBERT from 0.755 to 0.393, roughly a 47–48% relative drop. The models did not get worse; the measurement stopped being circular. These checkpoints had never been evaluated against labels their own dictionary annotator did not also produce, so silver scoring rewarded them for learning the dictionary's blind spots along with its correct patterns. For reference, the old GLiNER checkpoint (fine-tuned on the same dictionary-silver labels, not yet retrained on the current gold) drops by a smaller margin, 0.666 to 0.540 (19% relative): the asymmetry that motivated an initial, informal hypothesis for RQ4: that span-matching architectures generalize better than token classifiers once evaluation is no longer circular. Section 4.4 retrains GLiNER on the identical gold and shows that the smaller drop was not architectural: it reflects GLiNER's old training data happening to be less circular than the token classifiers' relative to *this specific independent evaluation set*, not an architecture effect that survives a controlled retrain.

The rescoring shows the size of the gap; a blind human audit establishes which side of the measurement was wrong. Three hundred predictions from the two baseline models were judged blind to model identity, each against the source text alone, with the sample deliberately stratified toward spans the silver evaluation had scored as false positives (205 of 300). Half of those silver-flagged "false positives" (103/205, 50.2%) were correct entities under direct human judgment (59% for the fine-tuned token classifier and 43% for zero-shot GLiNER), while the silver evaluation's true-positive verdicts were overwhelmingly confirmed (87% and 96%, respectively). The silver gold was not merely noisy: half of the errors it attributed to the models were entities it simply did not contain, which is precisely the failure mode a model trained on the same dictionary is rewarded for reproducing. The judged audit sheet, model-identity key, and scoring script are included in the released artifacts (Section 8).

4.2 RQ2: Scale vs. Domain Pretraining Under Independent Gold

Table 2 reports the three encoders after retraining on the LLM-gold training split and scoring against the same independent test gold. SecureBERT performs best (0.712 strict / 0.758 relaxed micro-F1), ahead of RoBERTa-base (0.659 / 0.683) and, notably, ahead of RoBERTa-large (0.648 / 0.697) despite RoBERTa-large having roughly 3× the parameters. Domain-adaptive pretraining recovers

Model	Circular F1	Indep. P	Indep. R	Indep. F1
RoBERTa-base	0.740	0.86	0.22	0.351
SecureBERT	0.755	0.87	0.25	0.393
GLiNER (v1, ref.)	0.666	0.65	0.46	0.540

Table 1: Checkpoints trained on the prior dictionary-silver pipeline, scored against the silver labels they were trained toward ("Circular") vs. against independent LLM+human gold ("Indep."), strict span-and-label match, 8-document test split.

Model	P	R	F1	F1 (relaxed)
RoBERTa-base	0.71	0.61	0.659	0.683
RoBERTa-large	0.71	0.60	0.648	0.697
SecureBERT	**0.73**	**0.70**	**0.712**	**0.758**
GLiNER	0.51	0.61	0.56	0.58

Table 2: Architecture comparison, identical LLM-gold training and evaluation labels, identical evaluation harness. Micro P/R/F1, 8-document test split, 1,995 gold spans.

Label	*n*	Base	Large	SecBERT
malware family	438	0.59	0.35	0.65
attack technique id	297	0.86	0.91	**0.99**
affected software product	159	0.48	0.57	0.61
threat actor group	157	0.29	0.42	0.34
attack technique name	144	0.57	0.53	0.57
hacking tool	139	0.37	0.44	0.51
ip address	130	0.85	0.85	0.92
tactic	111	0.59	0.63	0.62
file path	108	0.84	0.83	0.84
file hash	93	0.70	0.65	0.64
file name	89	0.76	0.81	0.74
cve identifier	55	0.92	0.96	0.94
tlp classification	36	0.97	0.97	0.97
email address	26	0.98	0.98	0.98
yara rule name	9	0.62	0.71	0.50
cwe identifier	4	1.00	1.00	1.00
Micro	1995	0.659	0.648	**0.712**

Table 3: Per-label strict F1, sorted by test-set support. Base = RoBERTa-base, Large = RoBERTa-large, SecBERT = SecureBERT.

more of the independent-gold quality here than raw scale does. RoBERTa-large's higher relaxed F1 and macro-F1 alongside a *lower* strict micro-F1 than RoBERTa-base is explained by the per-label pattern in Section 4.3: RoBERTa-large predicts the correct general region and type more consistently across labels (helping macro and relaxed matching) but regresses sharply on the single highest-volume label, *malware family*, which drags support-weighted strict micro-F1 down.

Comparing back to Table 1, retraining on independent gold closes most of the gap the circular evaluation had hidden. RoBERTa-base's score against independent gold rises from 0.351 (old checkpoint) to 0.659 (retrained), most of the way back to its old, circular 0.740, and SecureBERT's independent-gold score (0.712) nearly matches its old circular number (0.755). Correcting *both* the training labels and the evaluation labels recovers almost all of the quality the circular number implied was already present; correcting only the evaluation labels (Table 1) is what made the models appear to regress.

Once retrained on this same gold, GLiNER (0.56 strict / 0.58 relaxed) trails every token classifier, including RoBERTa-large. Its macro-F1 (0.62 strict) sits well above its own micro-F1, the same pattern RoBERTa-large shows above: reasonably consistent type coverage dragged down by weak performance on a few high-volume labels, most strikingly *ip address* (F1 0.03, recall 0.02; Section 4.4 discusses possible causes). This directly answers the question raised in the RQ1 discussion: the old GLiNER checkpoint's smaller circular-vs-independent drop (Table 1) was not evidence of an architectural generalization advantage. Retrained under the identical procedure applied to the token classifiers, GLiNER is the weakest of the four models on this task.

4.3 RQ3: Which Entity Types Remain Hard

Table 3 breaks the three retrained encoders down by label. Structured, pattern-shaped labels are near-solved regardless of frequency: *attack technique id*, *cve identifier*, *tlp classification*, and *email address* all score 0.92–1.00 F1 across all three models, and even the lowest-support label in the test set (*cwe identifier*, $n = 4$) reaches 1.00 on every model, although at that support level a single miss would swing the number by 0.25, so this particular result should be read as indicative rather than as evidence that CWE identification is solved.

The weakest labels are not simply the rarest ones. *Malware family* has the largest test-set support of any label ($n = 438$) yet tops out at 0.65 F1 (SecureBERT); *threat actor group* ($n = 157$) and *hacking tool* ($n = 139$) similarly lag well behind the structured labels despite moderate-to-high support. Frequency alone therefore does not explain the gap: these are open-vocabulary, context-dependent labels (whether a string names a piece of malware, a named intrusion set, or a legitimate administrative tool such as PowerShell often depends on surrounding sentence context rather than surface form), and that semantic ambiguity appears to be a harder problem for a token classifier than sparse supervision is. RoBERTa-large's regression on *malware family* specifically (0.35 vs. 0.59 for RoBERTa-base and 0.65 for SecureBERT, on identical training data) is the single largest per-label swing in the table and the main driver of RoBERTa-large's lower strict micro-F1 in Table 2.

4.4 RQ4: Architecture Comparison

GLiNER was fine-tuned on the identical LLM-gold training split used for the three token classifiers (5 epochs, learning rate 5e-5, effective batch size 32; best checkpoint by validation loss) and scored with the identical harness on the identical 8-document test split. Table 2's result is unambiguous: GLiNER (0.56 strict F1) is the weakest of the four architectures tested, behind RoBERTa-large (0.65), RoBERTa-base (0.66), and SecureBERT (0.71). This resolves the question RQ1 raised: the initial hypothesis does not survive controlled retraining. Section 4 sketches why: the old GLiNER checkpoint's

Label	GLiNER F1	Best token clf. F1
ip address	0.03	0.92 (SecBERT)
file path	0.42	0.84 (Base/SecBERT)
threat actor group	0.29	0.42 (Large)
hacking tool	0.38	0.51 (SecBERT)
malware family	0.47	0.65 (SecBERT)
file name	0.46	0.81 (Large)
tactic	0.56	0.63 (Large)
attack technique name	0.50	0.57 (Base/SecBERT)
attack technique id	0.99	0.99 (SecBERT)
cve identifier	0.96	0.96 (Large)
cwe identifier	1.00	1.00 (all)

Table 4: GLiNER strict F1 vs. the best-performing token classifier per label (selected labels; full 16-label GLiNER results in `data/gold_v2/results/trained-gliner-v2.json` in the project repository, Section 8).

Model	Strict F1	Relaxed F1	Avg. latency/doc
Gemini 3.5 Flash	**0.795**	**0.803**	94.5 s (7.2–538.7 s)
SecureBERT	0.712	0.758	**1.8 s** (0.4–5.2 s)
RoBERTa-large	0.648	0.697	7.8 s (1.1–30.1 s)
RoBERTa-base	0.659	0.683	n/a
GLiNER	0.560	0.580	59.7 s (12.3–171.6 s)

Table 5: All five architectures, identical LLM-gold evaluation, identical harness, 8-document test split. Latency is wall-clock document-level inference on identical hardware (Apple Silicon, no batching) for local models, and real API round-trip time for Gemini (Section 3.4): not directly comparable across the network/local boundary in a strict sense, but both are genuine end-to-end costs a deployment would pay. RoBERTa-base latency was not re-measured because its local checkpoint was not preserved (see text).

smaller circular-vs-independent drop reflected its old training data's relationship to *this* evaluation set, not a property of span-matching architecture that a fair retrain preserves.

Table 4 shows GLiNER is not uniformly weak: it matches or nearly matches the token classifiers on several structured labels (attack technique id 0.99, cwe identifier 1.00, cve identifier 0.96), but is markedly worse on *file path* (0.42 vs. 0.83–0.84 for the token classifiers) and severely degraded on *ip address* (0.03 vs. 0.85–0.92), a label none of the three token classifiers found difficult. We do not have a confident explanation for the *ip address* collapse. One contributing factor we can document, though not fully attribute, is that GLiNER handles its own tokenization and truncates at a 384-token limit rather than using the token classifiers' sliding-window scheme (Section 3): several of the longest test-set chunks were silently truncated during evaluation, which could plausibly cost recall on labels that cluster late in long chunks (indicator tables often appear near the end of CISA advisories). We have not verified that IP addresses specifically follow that positional pattern here, and we flag this as a concrete follow-up rather than a settled explanation.

4.5 RQ4 (continued): Frontier-LLM Comparison

Table 5 compares Gemini 3.5 Flash, run zero-shot with no fine-tuning, against the best fine-tuned local model (SecureBERT) on the identical test set and harness. Gemini wins on quality: 0.795 strict / 0.803 relaxed micro-F1, roughly 8 points above SecureBERT's 0.712 / 0.758, and macro-F1 shows the same ordering (0.791 vs. 0.738): the advantage is not an artifact of a few high-volume labels dominating a support-weighted average. This outcome is the inverse of the premise that motivated the fine-tuned model set (that a small fine-tuned model could match a frontier model on a narrow domain task), and we report it as measured.

The cost is real and asymmetric, and, unexpectedly, it is not evenly distributed across the local models. Table 5 measures wall-clock inference on the same 8 documents, on the same hardware (Apple Silicon, CPU/MPS, no batching), for every model in this paper. SecureBERT is not only the most accurate local model; it is also by far the fastest, averaging 1.8 s per document, 51× faster than Gemini, with no network call, no per-token billing, no rate limits, and no marginal cost once the checkpoint is downloaded. RoBERTa-large is a step down on both axes (0.648 F1, 7.8 s/document, 12× faster than Gemini). GLiNER, however, departs from the expected pattern: despite running entirely locally, it averages 59.7 s per document, only 1.6× faster than a cloud API call, and dramatically slower than either token classifier. Span-matching against a label-embedding space is architecturally more expensive per token than a per-token softmax head, and that cost is visible directly here: GLiNER is dominated on *both* accuracy and speed by SecureBERT, without the "less accurate but free and fast" trade-off that would make it a defensible choice on locality grounds. Whether Gemini's F1 advantage over SecureBERT justifies its cost depends entirely on deployment context (a security operations center triaging thousands of advisories per day cannot absorb cloud-API latency and cost at that multiplier, whereas a one-off deep analysis of a handful of critical advisories plausibly can), but for GLiNER specifically there is no such trade-off to weigh: SecureBERT surpasses it on every axis measured. Confidentiality sharpens the deployment argument further: the advisories studied here are public (TLP:CLEAR), but operational CTI routinely circulates under TLP:AMBER or TLP:RED handling restrictions that preclude submission to a third-party cloud API at any price. In such settings the local model is not the economical choice but the only admissible one, and the F1 gap is the cost of compliance rather than a trade-off to optimize. RoBERTa-base's original local checkpoint was not preserved for re-measurement (only its evaluation results were retained); given that its architecture and training recipe are identical to RoBERTa-large's, its latency is expected to fall in the same range but is not independently confirmed here.

Two pipeline defects surfaced and were fixed while producing this comparison; both are worth documenting because they illustrate the same lesson as RQ1: an unverified evaluation harness produces incorrect numbers with no indication of error. First, span extraction matched entity text against a 4000-character truncation window left over from an earlier, non-chunked inference mode; with full-document chunked inference, this silently dropped roughly 72% of Gemini's correctly identified entities before scoring ever saw them (732 raw entities → 203 matched spans). Corrected to match against the full document, spans recovered from 203 to 2,318 (gold,

like this pipeline, credits every occurrence of a repeated entity mention, not only the first). Second, frontier scoring used a parallel, independently written implementation of strict P/R/F1 instead of the shared `ner.evaluate.Scorer` used for every other model in this paper: mathematically equivalent once checked, but a maintenance hazard and a literal violation of this paper's own "identical scoring implementation across models" claim (Section 3.4). Both defects are fixed in the released pipeline; the numbers above reflect the corrected code.

Appendix A extends this comparison with a wider zero-shot sweep on the identical harness and test split: two further Gemini variants, a reasoning-mode ablation, Kimi K3, Claude Haiku 4.5, Claude Opus 4.8, and a locally hosted open-weights gemma3:27b. None of the additional models exceeds Gemini 3.5 Flash, so the comparison above is representative; the appendix material is complementary detail rather than a change to any conclusion here.

5 Limitations

Training gold is LLM-only, not human-verified. Human curation is applied to the 16 validation/test advisories but not to the 66 training advisories (Section 3), for cost reasons. If Claude Sonnet's proposals have systematic blind spots of their own (the same structural risk that motivated moving off the dictionary annotator, with a different annotator), training labels would inherit them even though the evaluation labels, being human-curated, would not. The results in Section 4 should therefore be read as "evaluated without circularity," not as "free of all annotator bias"; only the measurement side of that distinction is currently guaranteed.

No blind audit of the LLM-primary gold. The blind human audit in Section 4.1 covers baseline-era predictions and silver-gold verdicts; no equivalent model-identity-blind audit has been run over the current pipeline's gold or the retrained models' predictions. The current gold is human-checked only through the val/test curation pass, which is exhaustive over those documents but not blind. The audit protocol and tooling (`ner.audit_webapp`, `ner.score_audit`) transfer directly; applying them to the LLM-primary gold is future work.

Test set size. Eight held-out advisories (1,995 gold spans in the test set) is larger than the silver baseline's test set (1,442 dictionary-silver spans) and covers label types dictionary annotation structurally could not, but the lowest-support labels (*cwe identifier*, $n = 4$; *yara rule name*, $n = 9$) still have too few examples for their per-label numbers to be read as precise.

Windowing artifacts. All three token classifiers resolve each word's tag from whichever sliding window contains it most centrally at inference time, which is a heuristic rather than a principled ensembling method; a small fraction of predictions near window boundaries may be affected.

GLiNER truncation, not windowing. Unlike the token classifiers, GLiNER's own tokenizer truncates rather than sliding-windows long input, and several of the longest test-set chunks were truncated at its 384-token limit during evaluation. We do not currently know how much of GLiNER's per-label weakness (Section 4.4), particularly the sharp *ip address* drop, is attributable to this truncation versus a genuine architecture or data effect; disentangling the two (e.g., by re-chunking GLiNER's input at a shorter budget matching its limit, mirroring the token classifiers' windowing) is left to future work.

Frontier comparison: single sample, one prompt per model. The main text compares against one frontier model (Gemini 3.5 Flash) with one prompt; Appendix A widens this to seven additional zero-shot configurations (further Gemini variants, a reasoning-mode ablation, Kimi K3, Claude Haiku 4.5, Claude Opus 4.8, and a local gemma3:27b), none of which exceeds Gemini 3.5 Flash. Even so, every run is a single sample with a single prompt; we did not sweep prompt variants, temperature, or few-shot examples, and a different prompt could plausibly close some of the 8-point F1 gap. Cost and latency figures are similarly a single measurement (one API tier, one region, one day's network conditions), not a stable production benchmark across load or over time.

6 Conclusion

Evaluating against labels produced by the same annotator being improved upon is circular, and the gap it conceals is substantial: a checkpoint that scored 0.74–0.76 F1 against dictionary-silver gold scores 0.35–0.39 against independently produced gold (RQ1). Retraining on that same independent gold recovers most of the apparent quality (SecureBERT reaches 0.71 F1, nearly matching its own old circular number), showing that the earlier result was not spurious; it was measured against an unsuitable reference (RQ2). Domain-adaptive pretraining (SecureBERT) recovers more independent-gold quality than raw parameter count (RoBERTa-large), which notably regresses on the single highest-volume label rather than improving on it. Structured, pattern-shaped entity types are essentially solved by all three encoder architectures; open-vocabulary, context-dependent types (most visibly *malware family* and *threat actor group*) remain the hard core of this task regardless of training volume (RQ3).

The architecture question this work set out to answer (whether GLiNER's span-matching design generalizes better than a token classifier's softmax head once measured without circularity) is answered, and not in the direction the earlier, uncontrolled comparison suggested. Retrained on the identical independent gold and scored with the identical harness, GLiNER (0.56 F1) is the weakest of the four architectures tested, behind every token classifier including RoBERTa-large (RQ4). The apparent GLiNER advantage in Table 1 was an artifact of comparing a retrained token classifier against an *un*retrained span-matching model: a reminder that the same circularity discipline this paper applies to silver-vs-independent gold (RQ1) must be applied equally strictly to any two checkpoints being compared, or a training-data confound can masquerade as an architecture finding.

Extending the comparison to a frontier LLM completes the evaluation: zero-shot Gemini 3.5 Flash exceeds every fine-tuned local model on F1, including SecureBERT (0.795 vs. 0.712 strict), at the cost of 94.5 s mean latency per document and metered per-token API pricing against SecureBERT's 1.8 s, free, offline inference: 51× faster for 8 fewer points of F1 (RQ4). That trade-off is at least well characterized: a straightforward speed-and-cost-for-quality exchange whose resolution depends on deployment context, and

one that data-handling policy can decide outright: material under TLP:AMBER or TLP:RED restrictions cannot be sent to a third-party API at any price, leaving the local model as the only admissible option for a substantial class of operational CTI. GLiNER's position admits no such justification: at 59.7 s per document it forfeits nearly all of the local-deployment speed advantage (1.6× over Gemini, not 51×) while also trailing SecureBERT on F1, so on this task, measured this way, it is dominated outright rather than trading accuracy for speed. Producing this comparison rigorously also required catching and fixing two evaluation-harness defects (Section 4.4) that would otherwise have quietly misreported it: the same discipline RQ1 required of the silver-vs-independent gold comparison, applied to a frontier-vs-local one.

7 Future Work

The per-label results (Section 4.3) suggest a concrete next step for the weakest, open-vocabulary semantic types: a gazetteer of canonical names and aliases for *threat actor group* and *hacking tool*. Both types have curated, actively maintained alias inventories: MITRE ATT&CK group and software entries and MISP Galaxy threat-actor clusters each enumerate dozens of aliases per named intrusion set (e.g., APT28 / Fancy Bear / Sofacy / STRONTIUM), and part of the observed difficulty is plausibly alias variety rather than genuinely novel vocabulary. Two design constraints follow from this paper's own findings. First, the gazetteer must enter the system strictly as inference-side model input (lookup features, gazetteer-aware decoding, or candidate reranking), never as a labeling source: using the same resource to produce or correct gold labels would reintroduce exactly the circularity RQ1 quantifies. Second, scored against the existing independent gold, a gazetteer provides a measurable decomposition of the remaining gap on these labels into alias-coverage error, which lookup can fix, and contextual-disambiguation error (e.g., PowerShell as routine administration versus as a *hacking tool* in an intrusion narrative), which it cannot, and which would then be isolated as the residual hard core of the task.

A second next step is external validation on an independently annotated corpus. AnnoCTR [6] is well suited to this: 400 vendor-donated cyber threat reports with expert annotations for named entities and for MITRE ATT&CK tactics and techniques (including the explicit/implicit mention distinction), released under CC-BY-SA 4.0, a license most earlier CTI corpora lack. Because AnnoCTR's gold was produced by a human expert with no involvement from this paper's annotation pipeline, it offers exactly the kind of evaluation this paper's methodology calls for but could not perform internally: a test of whether models fine-tuned on LLM-proposed gold generalize to labels whose provenance is fully independent of any LLM. The label schemas do not align one-to-one, so a mapping layer would be required, and the cross-corpus NER literature cautions that such mappings are themselves lossy; measuring how much of the transfer gap is schema mismatch versus genuine generalization failure is part of the value of the exercise.

8 Artifact and Data Availability

All code, gold data, and evaluation artifacts required to reproduce the results in this paper are publicly available.

Code and data. The complete pipeline is available at https://github.com/dandye/nlp_capstone: the CISA scraper, text normalization, LLM annotation (`ner.llm_gold`), human-curation webapp (`ner.hitl_ner`), training (`ner.finetune_roberta`, `ner.finetune_gliner`), frontier comparison (`ner.frontier_infer` / `ner.frontier_extract` / `ner.frontier_score`), and the shared evaluation harness (`ner.evaluate`). Gold annotations are under `data/gold_v2/` (evaluation, human-curated) and `data/gold_v2_train/` (training, LLM-only); the exact train/validation/test partition is `data/gold_v2/split_full.json`. The per-model scores reported in every table are committed as JSON under `data/gold_v2/results/` (`trained-*.json` for models retrained on the current gold, `rescored-*.json` for the RQ1 circularity rescoring) and `data/frontier_results/` (Gemini entities, spans, and scores). The blind-audit artifacts backing Section 4.1 are likewise included: the judged sheet (`paper/capstone/gold_audit.csv`), the model-identity key, and the scorer (`ner.score_audit`), along with the judging webapp (`ner.audit_webapp`).

Models. The fine-tuned checkpoints are published on the Hugging Face Hub: https://huggingface.co/pzryathzsdhc/cti-ner-roberta-large, https://huggingface.co/pzryathzsdhc/cti-ner-securebert, and https://huggingface.co/pzryathzsdhc/cti-ner-gliner. A demonstration notebook, `notebooks/hf_models_demo.ipynb`, downloads the published checkpoints and reproduces windowed inference on sample advisory text; the Colab notebooks under `notebooks/` reproduce the training runs themselves.

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Model	F1	P	R	Macro	s/doc	Tokens
Gemini 3.5 Flash	**0.795**	0.739	0.859	0.791	94.5	350,277
Gemini 3.6 Flash (think.)	0.794	0.693	0.928	**0.811**	601.1	1,410,327
Kimi K3	0.761	0.652	0.913	0.784	1,118.3	434,025
Gemini 3.5 Flash-Lite	0.752	0.694	0.822	0.762	**30.7**	210,533
Gemini 3.6 Flash (min.)	0.707	0.564	**0.946**	0.773	68.5	231,260
Claude Haiku 4.5	0.699	0.623	0.797	0.678	50.8	190,290
Claude Opus 4.8	0.420	0.285	0.795	0.652	116.9	247,576
gemma3:27b (local)	0.240	0.146	0.690	0.379	3,114.0	89,909
SecureBERT (fine-tuned)	0.712			0.738	1.8	

Table 6: Extended zero-shot sweep: strict micro P/R/F1, macro F1, mean API round-trip latency per document, and total tokens recorded across the 8-document test split. Gemini token counts are `total_token_count` and therefore include hidden reasoning ("thinking") tokens. gemma3:27b ran locally via Ollama on Apple Silicon; its latency is local inference time, not a network cost. The fine-tuned SecureBERT row from Table 5 is repeated for reference.

A Extended Zero-Shot Frontier Comparison

The main text (Section 4.4) compares a single frontier model, Gemini 3.5 Flash, against the fine-tuned local models. This appendix reports a wider zero-shot sweep run on the identical pipeline (`ner.frontier_infer` → `ner.frontier_extract` → `ner.frontier_score`), identical 8-document test split, identical 4,000-character chunking, and identical shared scorer (`ner.evaluate.Scorer`). It is complementary material: none of the additional models exceeds Gemini 3.5 Flash, so every main-text conclusion stands without it. Per-model entities, spans, and scores are committed under `data/frontier_results/`.

Reasoning ("thinking") configuration. Gemini 3.5 Flash and 3.5 Flash-Lite ran with `thinking_budget=0`, reasoning fully disabled. Gemini 3.6 Flash rejects that setting, so it appears twice: once with provider-default reasoning enabled and once at `thinking_level="minimal"`, the lowest level its API accepts (documented as lowest, not necessarily zero, so the no-reasoning conditions across model generations are close but not strictly identical). The pair is an unplanned ablation with a clear result: default reasoning costs roughly 6× the tokens and 9× the wall-clock time and buys +0.087 strict F1, almost entirely as precision (0.693 vs. 0.564; recall is flat at 0.93–0.95). The damage concentrates in open-vocabulary labels (*affected software product* 0.62 to 0.37, *attack technique name* 0.82 to 0.65, *threat actor group* 0.60 to 0.51), consistent with reasoning acting as a candidate filter. Notably, 3.5 Flash reaches 0.795 with reasoning fully off, while 3.6 Flash needs default reasoning just to tie it.

Protocol caveats. Two models required deviations from the JSON-with-offsets output protocol the Gemini runs used, and their rows should be read with that in mind. *Kimi K3* (Moonshot API) returned empty responses for any structured-output request, so it emitted free-form entity/label lists that were located in the document by string search before the shared extraction step; its 1,118 s/document is API latency, not a statement about model depth. *Claude Haiku 4.5* carries a same-family caveat: the gold in this paper was proposed by Claude Sonnet 5 before human curation (Section 3), so a Claude-family model could plausibly share span-boundary and label-choice conventions with the annotator. Despite that potential tailwind it scores below fine-tuned SecureBERT, and it exhibits the largest single-label failure in the sweep: it emitted 1 *tactic* entity against 111 gold mentions (per-label F1 0.00). *Claude Opus 4.8* (same same-family caveat) illustrates a different hazard, a model-behavior interaction with the harness itself: its strict micro F1 of 0.420 (precision 0.285, recall 0.795, macro F1 0.652) is dominated by a handful of short generic strings it proposed as entities. On the Snake-malware advisory, which quotes C source code, it labeled the bare string "C" an *affected software product*; the extraction step, which credits every text occurrence of a proposed entity, multiplied that single judgment into 372 false-positive spans, with "Windows", "RDP", and "PowerShell" behaving similarly (754 deduped entities became 5,563 spans, versus 2,551 for Haiku). Every model passes through the identical extraction step; the others do not emit bare generic tokens. This is the same lesson the main text draws twice (Section 4.4): under an occurrence-crediting harness, a single generic proposal is amplified into hundreds of false positives, which primarily depresses micro-averaged precision.

Local zero-shot collapse. gemma3:27b, the one locally hosted open-weights model in the sweep, collapses to 0.240 strict F1 (precision 0.146) at 3,114 s per document. The comparison with SecureBERT (0.712 at 1.8 s) makes the deployment point from Section 4.4 concrete: what makes a local model competitive on this task is fine-tuning on independent gold, not local hosting per se; a local model without fine-tuning combines frontier-level latency with sub-baseline accuracy.

Cost. Costs are estimated from recorded per-run token counts at published list rates (paid tier, July 2026: Gemini 3.5 Flash $1.50/$9.00 per million input/output tokens, 3.6 Flash $1.50/$7.50, 3.5 Flash-Lite $0.30/$2.50, Claude Haiku 4.5 $1.00/$5.00; reasoning tokens bill at the output rate), not from account-level billing, which covered unrelated concurrent work and cannot be attributed to individual runs. Assuming an 80/20 input/output split, per-document estimates are roughly $0.13 for 3.5 Flash, $0.02 for Flash-Lite, $0.08 for 3.6 Flash at minimal reasoning, and $0.04 for Claude Haiku 4.5. The default-reasoning 3.6 Flash run is decomposed differently: both 3.6 Flash runs sent identical prompts, so the minimal run's input estimate (roughly 185K tokens) is taken as the reasoning run's input, leaving roughly 1.23M output-plus-reasoning tokens and a run cost near $9.50, or $1.18 per document, roughly 9× the cost of 3.5 Flash for a statistically indistinguishable F1. These are single-run estimates, not a pricing benchmark, but the reasoning multiplier is structural: reasoning tokens are metered at the output rate, so the roughly 6× token inflation flows directly into cost at any rate schedule.